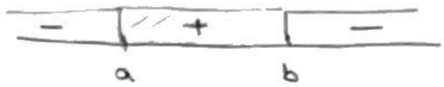
Δ Lecture 4 ⇝

* VC-Dimension of other hypothesis space ⇝

□ First thing that we should consider is shattering ⇝
for instances we have real numbers ($\mathbb{R}$) ⇒ $\frac{1}{=}$ intervals $[a,b]$ defines the positive spaces ⇝



$$h(x) = \begin{cases} +1 & \text{if } a \leq x \leq b \\ -1 & \text{otherwise} \end{cases} \quad \leadsto VC(H) = 2$$

we can shatter any dataset of two reals, but we cannot shatter dataset of three reals.

□ examples ⇝

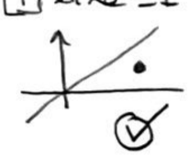
1. if arbitrarily large finite subsets of X can be shattered by X then $VC(H) = \infty$

2. if there exists one (or more) subsets of size-d that can be shattered, then $VC(H) \geq d$

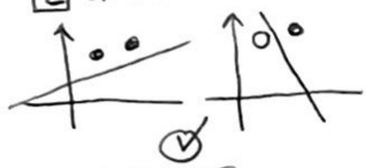
3. if no subset of size d can be shattered, then $VC(H) < d$

4. VC-dimension of linear classifiers in 2 dimensions:

[1] size = 1

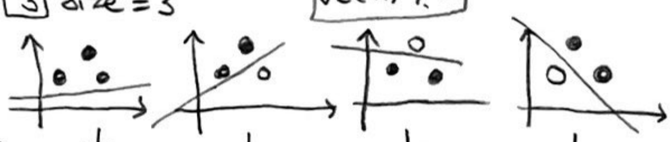


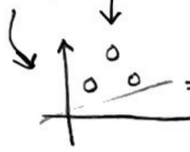
[2] size = 2



$\boxed{VC(H) \geq 1}$

[3] size = 3 $\boxed{VC(H) \geq 2}$



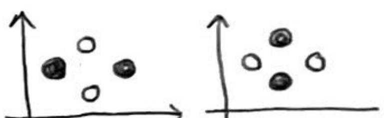


⇒ Just like other instances change the colors

[4] size = 4

⇝ all of the possibilities are accepted except this two situations :

$\boxed{VC(H) = 3}$



hypothesis class dimension	vc-dimension	parameters
1-D	1	θ
2-D	3	w_1, w_2, θ
interval	2	a, b

⇒ based on lectures

5. for finite H ⇝ $VC(H) \leq \log_2^{|H|}$

proof ⇝ a set S with d instances has 2^d distinct dichotomies. Hence H requires 2^d hypotheses to shatter d instances ⇝
if $VC(H) = d$ ⇝ hence :

$$2^d \leq |H|$$

$$VC(H) = d \leq \log_2^{|H|}$$